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REVIEW

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Exploring the impact of gamification on students' academic performance: A comprehensive meta-analysis of studies from the year 2008 to 2023

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Abstract

Gamification, characterized by the integration of game design elements into non-game environments, has gained popularity in classrooms due to its potential for increased engagement and enjoyment compared to traditional lecture-based teaching methods. While students generally exhibit positive attitudes towards gamification, its impact on academic achievement remains a subject of debate. This study employed a meta-analysis approach to examine the overall influence of gamification on students' academic performance. The sample comprised 22 experimental studies conducted between 2008 and 2023, comparing the effects of gamified and non-gamified classes. Utilizing a random effects model, the results revealed a moderately positive effect of gamification on student academic performance (Hedges's $g = 0.782$, $p < 0.05$). The paper further discussed the outcomes of various moderator analyses, providing valuable insights into the selection and utilization of game design elements, as well as considerations specific to different educational stages.

KEYWORDS

academic performance, gamification, meta-analysis

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Practitioner notes

What is already known about this topic

- Most research has consistently demonstrated that gamification has a positive impact on students' achievement.
- The current state of review research is not sufficiently comprehensive.
- There is a lack of meta-analyses exploring the diverse impacts of gamification.

What this paper adds

- The effect of factors such as geographical regions, education levels, learning environments, subjects and game elements on gamification was examined.
- The study revealed a significant and positive impact of gamification on students' achievement across various factors, including geographical regions, education levels, learning environments, subjects and game elements.

Implications for practice and/or policy

- Gamification represents a prudent choice for teachers seeking to enhance students' achievement.
- Teachers are suggested to adopt and employ appropriate game elements in their instructional approaches.
- Future research could focus on investigating the impact of feedback as a game element in teaching and learning.

INTRODUCTION

In recent years, the integration of e-learning platforms has become increasingly prevalent in educational institutions, acting as a modern complement or alternative to traditional teaching methods (Górska, 2016). The onset of the Covid-19 crisis has accelerated this shift, prompting higher education institutions to swiftly transition from face-to-face to online teaching to sustain student learning (Khaldi et al., 2023). Beyond the pandemic, the influence of e-learning endures, leading to broader adoption of online blended formats for university course instruction (Cobo-Rendón et al., 2022). However, the digital learning environment poses challenges, with students often experiencing reduced motivation and diminished engagement, resulting in decreased course participation due to distractions from external factors such as social media or household responsibilities (Barrot et al., 2021; Chiu et al., 2021; Suriagiri et al., 2022). Concurrently, educators grapple with maintaining learners' attention, potentially leading to disengagement from online learning (Denden et al., 2021). Moreover, students exhibit diverse learning styles and paces, posing a challenge for educators to meet individual needs and provide effective feedback (Tikhonova et al., 2022; Torres Martín et al., 2021).

Gamification, as a student-centred teaching method, holds promise in cultivating students' motivation and problem-solving abilities, providing a practical solution within the teaching process. Empirical research in higher education contexts supports gamification's positive impact on motivational outcomes, indicating that game design elements contribute to fulfilling competence and relatedness needs (Subhash & Cudney, 2018). Grounded in the self-determination theoretical framework, gamification is believed to foster intrinsic motivation while satisfying the needs for competence and social bonding Ryan et al., 2006). Specifically, by optimizing the learning process and enhancing learning outcomes, gamification plays a vital role in sustaining students' engagement and other dimensions of their

learning experience (Toda et al., 2019), foster learning persistence, facilitate communication (Zhao et al., 2022) and cultivate higher-order thinking skills among learners (Cheng et al., 2023).

Researchers have investigated the fusion of games with fields such as healthcare (Martinho et al., 2020), training, transportation and education (Hursen & Bas, 2019), yielding remarkable findings in academic achievement and skills development (De Freitas, 2018). A notable application has been the convergence of education and gaming, giving rise to terms like educational games, serious games and gamified learning (Kirchner-Krath et al., 2021). All these terms share the objective of enabling students to learn through gameplay by incorporating game elements or features that stimulate their interest (Ibanez et al., 2014) and immerse them in the learning experience (Tsay et al., 2018). The existing body of literature on gamification encompasses diverse perspectives and objectives, offering valuable insights into its impact on achievement. However, integrating these multidisciplinary findings and discussing them within a unified framework poses a significant challenge. Traditional literature reviews have been utilized in previous research to classify existing studies and provide an overview of the research landscape (Oliveira et al., 2023). Nevertheless, they often lack comprehensive integration and innovative analysis of the research outcomes (Riar et al., 2022).

To bridge the identified gap, this study employs a meta-analytical review approach, an emerging methodology, to amalgamate and synthesize current research findings on gamification, aiming to deliver a comprehensive and innovative analysis of the research in gamification. Through this approach, the study seeks to generate new knowledge and enrich our understanding of the implications of gamification in education.

LITERATURE REVIEW

The definition of gamification

The concept of gamification, which involves using game elements to engage and motivate individuals, was first mentioned in a 2002 paper by Nick Pelling titled 'The Chemistry of Game Design' (Hebebci & Selahattin, 2021). However, it was not until 2008 that the term 'gamification' was coined by Bret Terrill and gained widespread recognition in academia and industry (Terrill, 2008). Since then, gamification has increasingly been recognized as a powerful tool to motivate individuals in various contexts, including education, health and the workplace (Chapman et al., 2023). Gamification utilizes game elements to enhance user engagement, productivity and enjoyment (Deterding et al., 2011; Petrovych et al., 2023). By tapping into individuals' intrinsic desires for competition, achievement, recognition and self-expression, gamification has become a popular strategy for organizations worldwide. Many organizations are now adopting gaming technology and implementing playful rewards to motivate both employees and customers.

Gamification involves applying game elements to non-game activities, creating an immersive and interactive experience. It should be distinguished from game-based learning (GBL), which integrates complete, pre-designed games into the learning journey to provide a comprehensive teaching approach for subjects or specific modules (Dichev & Dicheva, 2017). While GBL utilizes complete gameplay, gamification applies game elements to non-game contexts. It is also important to note that gamification is different from learning-based games, as it encompasses the entire learning process transformed into a game-like experience (Kirchner-Krath et al., 2021). Importantly, gamification does not replace the learning process but rather enhances it by fostering active participation and encouraging learners to overcome challenges over time.

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Furthermore, it is important to note that serious games constitute a subset of games specifically designed for purposes beyond pure entertainment (Cowan & Kapralos, 2017). Serious games have broad applicability for addressing diverse problems or objectives due to their customizable nature (Landers, 2014). It is crucial to distinguish between game-based learning and serious games. Game-based learning refers to a learning methodology, whereas a serious game is a tangible product that facilitates game-based learning (Becker, 2021). The definitions of serious games provide additional insights. According to game designers Michael and Chen (2005), a game's primary purpose extends beyond mere entertainment. Zyda (2005) offers a more precise definition, describing serious games as 'psychological competitions following specific rules, using entertainment to advance objectives in government, corporate training, education, health, public policy and strategic communication'.

Through the application of gamification, learning becomes a more immersive and engaging activity, inspiring individuals to overcome learning obstacles with dedication. By leveraging game elements, organizations can tap into individuals' intrinsic motivations, creating a more enjoyable and effective learning experience.

Relevant studies of gamification in education

Numerous empirical studies have explored the integration of gamification in education, consistently revealing its positive influence on learning outcomes. For instance, Gari et al. (2018) demonstrated a significant enhancement in student learning and engagement in computer science courses through gamification, surpassing traditional instructional formats (Landers, 2014). Similarly, Ortiz Rojas et al. (2017) observed heightened engagement among students learning programming through gamified approaches. However, conflicting findings have emerged regarding the effectiveness of gamification in educational contexts. Instances where gamification did not lead to improvements in student performance were highlighted by Almeida et al. (2023) and Koivisto and Hamari (2019). Zainuddin et al. (2020) even identified potential negative effects on motivation and engagement .

In response to the variability in outcomes, scholars have conducted systematic literature reviews and meta-analyses to establish more reliable conclusions. Bai (2019) focused on computer science, information science and engineering education, confirming gamification's positive impact within these domains. Zhan et al. (2022) emphasized the greater efficacy of gamification in text-based programming over graphical programming. Li's (2023) meta-analytic study compared gamification effects across disciplinary categories, revealing a stronger impact on science and technology disciplines than on liberal arts disciplines. Sailer and Homner's (2020) systematic synthesis demonstrated that empirical studies support gamification as an effective instructional method, particularly in enhancing behavioural learning outcomes, encompassing technical skills, motor skills or competencies. However, a common limitation observed in other studies, such as Yıldırım and Şen (2021), is the lack of specification regarding the methods used to measure learning outcomes.

In comparison with Kim and Castelli's (2021) study, the present research refined experimental subject categorization by distinguishing between elementary school, middle school, high school and university levels. Another study by Huang et al. (2020) focused on students' learning outcomes, reporting a small to medium effect size ($g = 0.464$) on clearly defined cognitive, affective and behavioural measures. However, like other studies, it did not specify the measurement methods for learning outcomes.

These studies underscore the need for comprehensive examinations of gamification's impact on education, considering different subject areas and instructional contexts. Synthesizing and analysing findings across disciplines can contribute to a more nuanced understanding of the effects of gamification on student learning. The research gap lies in

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the inconsistent methodologies used to measure learning outcomes in existing studies, warranting further exploration for a more comprehensive understanding of gamification's impact.

Review studies of gamification in education

A systematic review study conducted by Dicheva et al. (2015) provides a comprehensive overview of the existing literature on the use of gamification in education. The study analysed 173 articles published between 2010 and 2014 and categorized them based on educational context, gamification techniques, target learners and learning outcomes. The findings revealed a focus on higher education and STEM subjects, with points, badges and leaderboards being the most commonly employed gamification techniques. However, the study also identified gaps in the literature, such as a lack of research in K-12 education and non-STEM disciplines, highlighting the need for more rigorous empirical studies on the effects of gamification in education. Further evidence supporting the positive impact of gamification on student engagement, motivation, self-confidence, attitude, perceived learning and performance comes from the work of Subhash and Cudney (2018). Their findings were aligned with the use of game elements such as points, badges, leaderboards, levels, feedback and graphics in higher education settings. Moreover, gamification has been shown to enhance critical thinking and problem-solving skills, as students apply newly acquired concepts through integrated learning approaches (Yasin & Abbas, 2021).

In science education, a systematic review by Kalogiannakis et al. (2021) confirmed the effectiveness of gamification in increasing student motivation, engagement and learning outcomes. The authors emphasized the importance of aligning the design of gamification elements with learning objectives and learner characteristics while also advocating for more empirical research on the long-term effects of gamification in this domain.

A meta-analysis conducted by Nadi-Ravandi and Batooli (2022) examined the impact of gamification on student academic performance across various educational settings. Analysing 53 meta-analyses and systematic reviews published between 2010 and 2020, the study found a moderate to strong positive effect of gamification on academic achievement. Factors such as the type and level of gamification, intervention duration and student demographics were identified as potential influences on the effectiveness of gamification. In the context of programming education, a meta-analysis by Zhan et al. (2022) synthesized data from 22 studies and concluded that gamification significantly enhances students' programming performance. The study highlighted the effectiveness of gamified learning, longer intervention durations and Asian-based studies. Additionally, the findings indicated that the impact of gamification was consistent across different assessment types and age groups.

Collectively, these studies support the notion that gamification can have a positive impact on learning outcomes, motivation, engagement and other relevant factors in education. Considering various factors, such as the specific gamification techniques employed, learner characteristics, subject areas and intervention durations, can further optimize the effectiveness of gamification strategies.

RESEARCH PURPOSE AND QUESTIONS

By considering various perspectives and influencing factors, the study aimed to address the inconsistencies present in the current empirical research on the effectiveness of gamification. In addition, this study seeks to provide a systematic review and analysis of the existing literature, synthesizing the findings to offer a more comprehensive understanding of the

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impact of gamification on academic performance. This study mainly answered the following questions:

1. Does gamification have a positive effect on academic performance?
2. Does the effect of gamification differ in different geographical regions?
3. Does the effect of gamification differ in different learning environments?
4. Does the effect of gamification differ among different levels of education?
5. Does the effect of gamification differ when using different game elements?
6. Does the effect of gamification differ in different subjects?

METHODS

This review study combines the use of systematic literature review and meta-analysis to conduct a comprehensive analysis. The systematic literature reviews adhere to a scientific and rigorous methodology, employing a clear and replicable research process, making them an essential approach in educational research (Xiao & Watson, 2019). By focusing on the 'literature' as the research object and employing a pre-defined coding list, systematic literature reviews ensure reliability and objectivity through comprehensive search strategies and strict screening processes. The meta-analysis, as a systematic data aggregation technique, integrates and reanalyses the results of multiple independent studies on the same research topic (Waddington et al., 2012). Calculating a combined effect size enables the exploration of inconsistent findings and the estimation of the true correlation between variables (Bearman et al., 2012). In this study, meta-analysis is predominantly used to investigate the impact of gamification on learning outcomes. The systematic procedures for the search, identification, screening, coding and extraction of publications are detailed in the following sections.

Article search strategy

In this study, a set of search databases was carefully selected based on their performance and relevance in the fields of information science, computer science and education. ACM Digital Library, IEEE Xplore, ScienceDirect, Wiley Online Library, Scopus and Web of Science were chosen due to their recognized strength in these areas (Cavacini, 2015). Additionally, we conducted a systematic literature search across two academic databases, ProQuest and Taylor & Francis Online, as they have been recommended in the field of education research (Snyder & Snyder, 2008). This approach ensured a comprehensive coverage of relevant publications for inclusion in the study, considering that gamification has been applied in various fields for educational purposes.

To construct effective search queries, we utilized Boolean search strings that combined specific terms related to gamification, education, students and effects. The gamification-related terms included 'gamification', 'gamify', 'gamified' and 'gamifying'. For education, we used terms such as 'assessing', 'assessment', 'impact', 'effect', 'evaluating', 'evaluation', 'comparison', 'influencing', 'enhancing', 'outcome', 'result' and 'influence'. The term 'student' was covered by including variations such as 'learner', 'user' and 'participant'. Lastly, we focused on the effects of gamification by incorporating terms like 'learning', 'comprehension', 'satisfaction', 'motivation', 'achievement', 'performance', 'understanding', 'perception', 'efficacy', 'self-efficacy', 'confidence' and 'expectancy'. It's worth noting that the wildcard character '*' was not used in search terms to ensure compatibility with all databases, as some may not accept it.

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Inclusion and exclusion criteria

Inclusion criteria

The following inclusion criteria were applied in this study:

1. Non-literary domain focus: The study should primarily focus on non-literary domains, specifically information science and computer science.
2. Timeframe: The literature search encompassed studies published from 2008 to the present 2023.
3. Educational context: The study explicitly targeted students across different educational levels in formal education settings.
4. Comparative approach: The study should compare gamification with non-gamification class (traditional class), using a between-subjects design.
5. Research design: The study should employ a quasi-experimental or experimental research design.
6. Learning outcomes: The primary focus of the study should be on student learning outcomes, particularly test scores.
7. Language: The study must be written in English.

Exclusion criteria

The following exclusion criteria were applied in this study:

1. Data measured by scores: Only studies that measured data through scores were included. This criterion was used to ensure consistency in the measurement of academic achievement. Consequently, studies that utilized partial experiments classifying students were excluded (Kam & Umar, 2022).
2. Retrospective studies: Retrospective studies were excluded from the analysis due to their inherent limitations in controlling variables and potential issues regarding the reliability of results.
3. Indirect effects of gamification: The focus of this paper was on exploring the direct effects of gamification on academic achievement. Therefore, empirical studies that primarily investigated the indirect effects of gamification, such as examining the impact of gamification on classroom engagement and subsequently linking engagement to academic performance through regression analysis, were excluded (Cakiroglu et al., 2016).

Figure 1 shows a PRISMA flowchart illustrating the search process, identification, screening, coding and extraction steps, following the guidelines outlined by Moher et al. (2009).

Coding methods

In order to improve the reliability of the study's conclusions, subgroup analysis was conducted to account for the high heterogeneity present in the data. Subgroup analysis, also known as moderating variable analysis, allows for a more comprehensive examination of the research findings. In this study, contextual factors were considered as the basis for defining subgroups (Borenstein et al., 2009). These factors include the region, subject, education level, learning environment, experimental cycle and application of game elements. By analysing the impact of gamification within different contextual settings, a more nuanced

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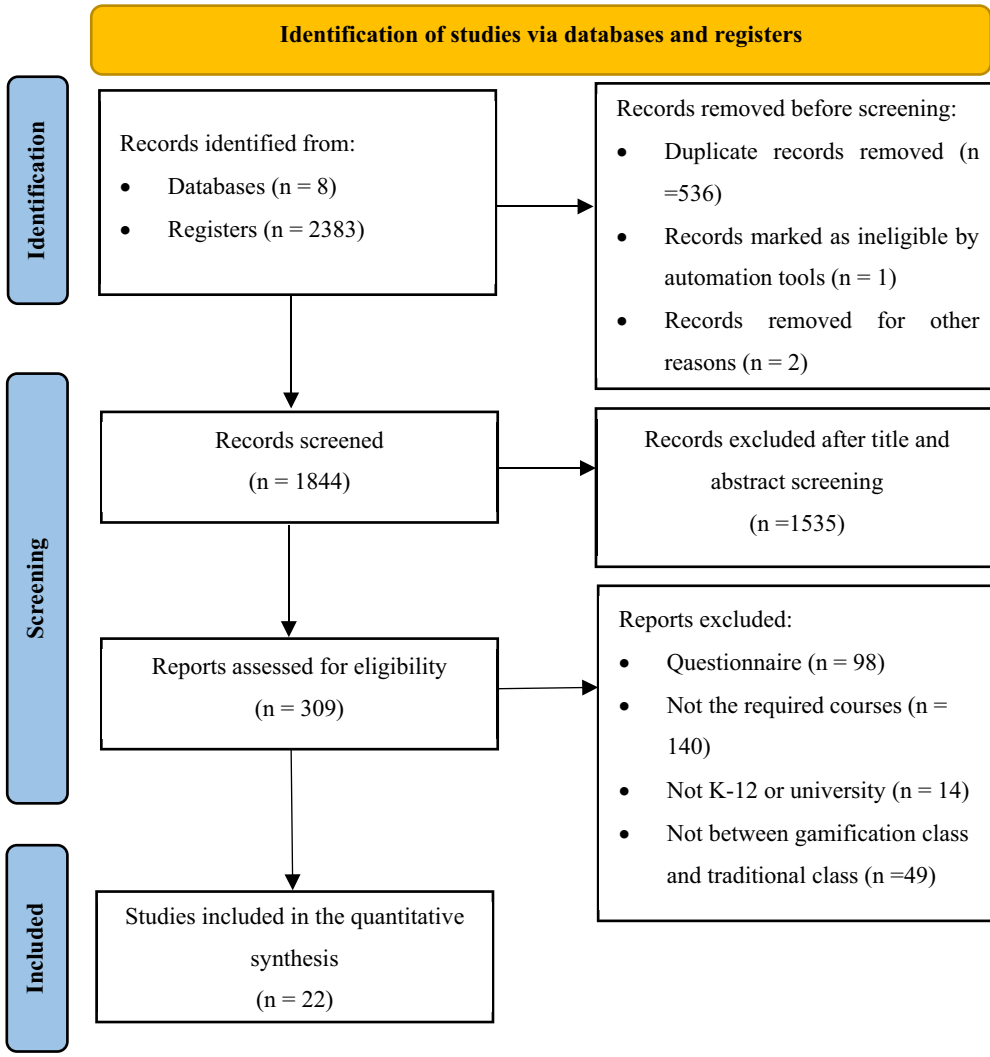

FIGURE 1 PRISMA flowchart of the article selection process.

understanding of its effects can be obtained. Finally, this study conducted a subgroup analysis based on five variables: region, subject, education level, learning environment and application of game elements.

In this study, the geographical ***regions*** comprised four categories based on the location of the experimental subjects. Regional disparities encompass diverse cultural backgrounds, educational values and teaching systems, all of which can potentially influence the effectiveness of gamification on academic performance. Different ***subjects*** exhibit distinct learning requirements, varying content difficulty levels and differing levels of student motivation, which may result in varied responses to gamification strategies (Yang et al., 2020). By examining the impact of gamification in various subject areas, researchers can uncover valuable insights into the optimal implementation of gamification strategies in different educational contexts. The ***education level*** encompasses elementary school, middle school, high school and undergraduate levels. The choice of educational level as a moderator factor is rooted

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in the understanding that different educational levels possess distinct needs, expectations and abilities, necessitating diverse game design strategies and mechanics (Nadi-Ravandi & Batooli, 2022; Ritzhaupt et al., 2021; Wu et al., 2021). The mode of instruction, whether online or face-to-face, may interact with gamification in distinct ways. By examining the effects of gamification in different ***learning environments***, researchers can enhance their understanding of how to effectively integrate gamification into various teaching contexts (Lim et al., 2021).

The selection and design of game mechanics in gamification applications depend on the specific context and objectives of the system (Subhash & Cudney, 2018). Several game elements are commonly utilized in gamification, including feedback, leaderboards, points, badges, challenges and immediate responses (Kirchner-Krath et al., 2021). In this study, the following game elements were identified and categorized according to related literature (Deterding et al., 2011; Hellín et al., 2023):

1. ***Points***: Points serve as a fundamental component of gamification. Students earn points upon successfully completing tasks or reaching milestones, reinforcing their progress and achievements.
2. ***Badges***: Badges are earned spontaneously after a certain period or upon accomplishing specific goals. The opportunity to collect badges and exchange them for rewards serves as a powerful motivator for students.
3. ***Leaderboards***: Leaderboards provide visual representations of individuals' achievements, enabling players to assess their performance and compare themselves to others. Leaderboards can be absolute, displaying rankings of top performers or relative, customized for each player and showing their position relative to a selected group of peers.
4. ***Challenges***: Challenges or tasks represent engaging and attainable objectives within the gamification framework. By presenting students with challenging yet achievable tasks, challenges serve as strong motivators for learning.
5. ***Feedback***: Immediate feedback/response is a crucial aspect of gamification. Students receive instant feedback that highlights their mistakes or areas for improvement, allowing them to make immediate corrections and enhance their learning experience.

By understanding the different game elements and their roles in gamification, educators and designers can make informed decisions when implementing gamified systems. The careful selection and integration of game mechanics can enhance student engagement, motivation and overall learning experience.

Data analysis

Heterogeneity

All reported *p*-values were two-tailed, and the *t*-test was used to assess heterogeneity. Despite having strict inclusion and exclusion criteria during the literature screening stage, ensuring the consistency of the included studies can be challenging. Therefore, it is important to conduct a heterogeneity test for each study before calculating the combined effect size. The results of the heterogeneity test served as a basis for model selection and fitness analysis. In this study, forest plots and *Q*-tests were employed to assess heterogeneity (Stogiannis et al., 2023).

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Effect size

To account for variations across different study implementations, we employed a random effects model to calculate effect sizes (Raudenbush, 2009). The reported effect size used in this study was Hedges's g, which represents the adjusted standardized mean difference between the two groups. This measure is particularly suitable for meta-analyses involving studies with varying sample sizes (Korpershoek et al., 2016). To ensure independence assumptions were met, effect sizes were calculated for each study based on independent samples of students. When a study reported multiple assessments for a single subject, we followed the recommendation of Freeman et al. (2014) and selected the most comprehensive assessment, such as the final exam, over other assessments. If a study reported multiple assessments at different time points (eg, Time 1, Time 2, …), or multiple tests without providing the most comprehensive assessment of overall results for the same group of participants, we used the Integrated Meta-Analysis software to calculate a single combined mean effect size.

For this study, the dependent variable 'student achievement' was treated as a continuous variable, and two types of data extraction were conducted: (a) mean, standard deviation and sample size for two-group control studies, and (b) mean, *t*-value and sample size for single-group pre/post-test studies. It has been observed that Cohen's d exhibits positive bias, resulting in a biased estimation (especially with small samples, where the error is greater). To address this, Hedges's g value was introduced as a correction factor (Borenstein et al., 2009). The calculation of Hedges's g is as follows, based on the mean, standard deviation and sample size (Hedges, 1981):

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}}$$

$$g = \frac{\overline{y}_1 - \overline{y}_2}{s_p}$$

where denotes the mean of the experimental group, denotes the mean of the control group, S_p denotes the combined standard difference, n_1 is the sample size of the experimental group, n_2 is the sample size of the control group, s_1 is the standard deviation of the experimental group and s_2 is the standard deviation of the control group.

Publication bias

As meta-analysis involves secondary analysis of existing studies, it is crucial to examine publication bias before conducting specific analyses to ensure the reliability of the results. Publication bias is an inherent issue in the meta-analysis, referring to the tendency of statistically significant studies to have a higher likelihood of being accepted and published compared to statistically insignificant studies (Spector & Thompson, 1991). In this study, we utilized comprehensive meta-analysis (CMA) to analyse publication bias, funnel plots, trim-and-fill analysis, rank correlation and linear regression were employed to assess potential bias in the included literature. The funnel plot is the most intuitive detection method, which judges whether there is bias by observing the distribution of the effect size included in the study on both sides of the centreline.

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RESULTS

Distribution of the empirical studies

Despite the slight impact of Covid-19, research in this field is generally experiencing an upward trend, as depicted in Figure 2. In recent years, there has been significant growth in gamification research. This is primarily due to the expanding popularity and wide-ranging applications of gamification. Scholars and researchers are increasingly recognizing the potential value of gamification in enhancing people's behaviour, improving efficiency and fostering motivation for participation. Gamification remains a prominent and relevant topic in contemporary discussions.

Description of the effect of gamification on students' achievement

Table 1 presents an overview of the selected articles concerning authors, the publication year, education level, subject, learning environment, gamification element(s) and region.

Distributions of studies according to the moderating variables

The study's moderator variable analysis revealed notable patterns across regions, education levels, game elements, subjects and learning environments (Table 2). With a predominant focus on Asia and Europe, the research indicates a growing interest in gamification's impact on undergraduate education, particularly in fields such as Computer Science. The gamification strategies employed varied widely, with a predominant emphasis on points, badges and leaderboards, often combined to enhance engagement and motivation. While subjects like Science and Computer Science garnered significant attention, other disciplines such as Education and Language also saw exploration. Furthermore, the study highlighted the diverse learning environments in which gamification was implemented, including blended, offline and online settings, with implications for educational practices and future research endeavours.

Random effects model

From Table 3, it is shown that the Q value is 349.066, with df = 21 and $p = 0.000$. Referring to the critical values of the chi-square distribution, the corresponding value for a 95% confidence interval with 21 degrees of freedom is 93.984. From the table of critical values for the

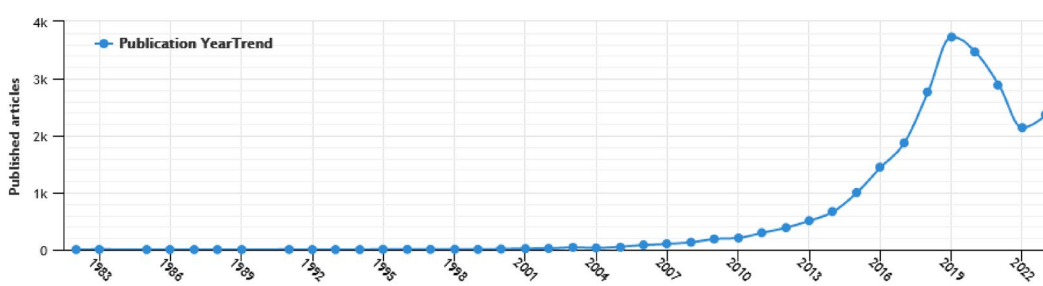

FIGURE 2 Number of publications per year.

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TABLE 1 Description of the use of gamification.

Authors (year)	Educational level	Subject	Learning environment	Gamification element	Region
Agapito and Rodrigo (2018)	Elementary	CS	Offline	Points+Badges+Leaderboard	Asia
Al Khateeb (2019)	Undergraduate	CS	Offline	Points+Badges+Feedback	Asia
Aljraiwi (2019)	Elementary	Science	Offline	None	Asia
Alsadoon et al. (2022)	Elementary	Language	Offline	Points+Badges+Leaderboard	Asia
Al-sulami (2022)	Undergraduate	Management	Offline	Points+Badges+Leaderboard	Asia
Carrasco-Hernández et al. (2020)	Undergraduate	CS	Blended	Level+Points+Badges	Europe
De-Marcos et al. (2014)	Undergraduate	CS	Offline	Points+Badges+Leaderboard	Europe
Elshemy (2017)	Undergrade	Education	Blended	Level+Badges	Europe
Gündüz and Akkoyunlu (2020)	Elementary	Science	Offline	Level+Badges	Asia
Karamert and Vardar (2021)	Undergraduate	CS	Blended	Level+Award+Badges+Leaderboard	Asia
Marin et al. (2018)	Middle	Science	Offline	Points	South America
Morales-Trujillo and García-Mireles (2020)	Undergraduate	Education	Blended	Points+Badges+Level+Leaderboard	North America
Ortiz Rojas (2017)	Undergraduate	CS	Blended	Level+Points+Badges	Europe
Rachels and Rockinson-Szapkiw (2017)	Highschool	Science	Offline	None	Europe
Sahin and Namli (2016)	Undergraduate	CS	Offline	Badges+Leaderboard	North America
Bernik et al. (2017)	Undergraduate	CS	Blended	Badges	Europe
Tasadduq et al. (2021)	Undergraduate	2016CS	Offline	Badges+LeaderBoard	Asia
Taskin and Kilic Cakmak (2022)	Elementary	Science	Offline	Points	Asia
Turan et al. (2016)	Highschool	CS	Offline	Points+Badges+Level+Leaderboard	Asia
Türkmen and Soyba (2019)	Middle	CS	Offline	Points+Badges+Leaderboard+Feedback	Asia
Wardoyo et al. (2021)	Undergraduate	CS	Blended	Points+Leaderboards	Asia
Yildirim (2017	Elementary	Language	Offline	Points+Badges+Leaderboard	Europe

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TABLE 2 Moderators in the current study.

Moderator variables	Categories	Studies
Region	Europe	7
	Asia	12
	South America	1
	North America	2
Education level	Elementary school	6
	Middle school	2
	High school	3
	Undergraduate	11
Game elements	Points	2
	Badges	1
	Points+leaderboards	1
	Levels+badges	2
	Levels+points+badges	1
	Badges+leaderboards	2
	Points+badges+leaderboards	5
	Points+badges+feedback	1
	Points+badges+leaderboards+feedback	1
	Points+badges+levels+leaderboards	2
	Levels+awards+badges+leaderboards	1
	No data	3
Subject	Education	2
	Science	5
	Language	2
	Computer Science	11
	Management	1
	Economy	1
Learning environment	Blended	6
	Offline	10
	Online	6

TABLE 3 Random effects model.

		95%CI		Heterogeneity			
Model	Hedges's g	LL	UL	*Q*	df	I^2	*p*
Fixed	0.727	0.669	0.786	349.066	21	93.984	0.000
Random	0.782	0.474	1.090				

chi-square distribution, $Q = 349.066$ is greater than 42.796, which confirms the presence of heterogeneity in the study. The *Q* value is influenced by the number of included studies but does not provide information about the magnitude of heterogeneity.

To assess the degree of heterogeneity, I^2 values were employed. These values indicated the overlap in confidence intervals of effect sizes and were independent of the size and distribution of true effects. I^2 values categorized heterogeneity as low, medium or high,

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corresponding to 25%, 50% and 75% increments respectively. In this study, $I^2 = 93.984$, which exceeds 75%, indicating a high level of heterogeneity among the original studies. Depending on the results of heterogeneity detection, statistical models could be selected. Statistical methods for meta-analysis included fixed-effects models and random-effects models. When heterogeneity is high, the random-effects model has more explanatory power for the results. Therefore, in this study, the results of the random effects model were chosen.

Overall effect of gamification on student learning outcomes

Figure 3 presents the forest plot, which provides a visual representation of the study's findings. The horizontal lines indicate the upper and lower limits of the confidence intervals, while the midpoints of the lines represent the effect values of each study. The size of the squares above the lines corresponds to the sample size, and the vertical line in the centre represents the no-effect line. The diamond below represents the combined effect values and their corresponding confidence intervals. Analysing the forest plot, it is evident that there is minimal overlap in the confidence intervals of individual studies, indicating the presence of heterogeneity among the studies. A total of 22 papers were included in this study, with a total sample size of 3144. According to the effect size measure proposed by Cohen (1992), the combined effect size had a small effect near 0.2, a moderate effect near 0.5 and a significant effect near 0.8. Twelve of the included studies reported significant effects, two reported moderate effects, seven reported low effects and two had negative effect values. The overall combined effect value as shown in Figure 3 (ES = 0.782), which lies between 0.4 and 0.8, implies that the use of gamification had a positive effect on learning outcomes, with a moderate to high effect.

Publication bias

Upon examining the funnel plot in this study, it was observed that the majority of included studies were concentrated in the middle and upper part of the funnel, displaying an even

Standard error	Variance	Study name	Time point	Hedges's g	Lower limit	Z-Value	Upper limit	p-Value
0.204	0.042	Yildirim (2017)	Blank	0.466	0.065	2.280	0.866	0.023
0.323	0.104	Elsherry (2017)	Blank	1.978	1.345	6.124	2.611	0.000
0.154	0.024	Rachels & Rockinson-Szapkiw (2018)	Blank	-0.052	-0.355	-0.339	0.250	0.734
0.173	0.030	Alsadoon et al. (2022)	Blank	0.081	-0.257	0.471	0.419	0.637
0.232	0.054	Turan et al. (2016)	Blank	0.499	0.045	2.152	0.953	0.031
0.409	0.167	Agapito & Rodrigo (2018)	Blank	0.081	-0.720	0.198	0.882	0.843
0.428	0.183	Al Khateeb (2019)	Blank	4.049	3.210	9.454	4.889	0.000
0.301	0.091	Taskin & Kilic Cakmak (2022)	Blank	0.774	0.184	2.573	1.364	0.010
0.292	0.085	Tasadduq et al. (2021)	T1	0.180	-0.392	0.616	0.751	0.538
0.199	0.040	Ortiz Rojas (2017)	Blank	0.257	-0.133	1.292	0.648	0.196
0.185	0.034	Carrasco-Hernández et al. (2020)	Blank	1.255	0.894	6.800	1.617	0.000
0.285	0.081	Strmecki et al. (2015)	Blank	1.064	0.506	3.737	1.621	0.000
0.355	0.126	Al-sulami (2022)	Blank	2.304	1.608	6.488	3.000	0.000
0.166	0.028	De-Marcos et al. (2014)	Blank	-1.254	-1.579	-7.555	-0.929	0.000
0.282	0.079	Türkmen & Soyba (2019)	Blank	0.284	-0.268	1.008	0.836	0.314
0.297	0.088	Karamert & Vardar (2021)	Blank	0.619	0.037	2.084	1.201	0.037
0.104	0.011	Marin et al. (2018)	T1	0.382	0.178	3.666	0.587	0.000
0.237	0.056	Gündüz & Akkoyunlu (2020)	Blank	0.684	0.220	2.889	1.148	0.004
0.445	0.198	Sahin & Namli (2016)	Blank	0.752	-0.119	1.693	1.624	0.091
0.231	0.053	Morales-Trujillo & García-Mireles (2020)	Blank	1.376	0.923	5.956	1.829	0.000
0.291	0.085	Aljraiwi (2019)	Blank	1.534	0.964	5.275	2.104	0.000
0.038	0.001	Wardoyo etal. (2021)	Blank	0.896	0.820	23.277	0.971	0.000
0.030	0.001			0.727	0.669	24.314	0.786	0.000

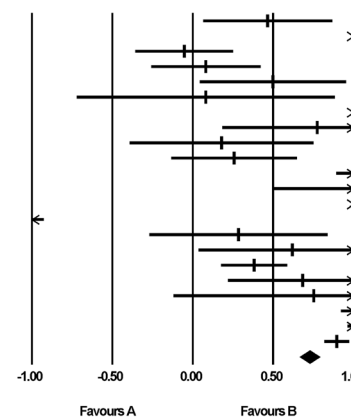

FIGURE 3 The forest plot of effect sizes (Hedges's g) using the random-effects model.

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distribution on both sides of the centreline (Figure 4). Only a few studies exhibited slight deviations from the centreline. Based on this preliminary analysis, it can be inferred that no significant publication bias exists in this study.

In this study, the results of Egger's test indicate $p = 0.384$ (>0.05), which suggests no significant publication bias. On the other hand, Begg's test yields $z = 2.14$ (>1.96). However, considering the relevant information and the suitability of Egger's test for small sample analysis, Egger's results were deemed more accurate and were therefore chosen. Consequently, the conclusion was that there was no publication bias in this paper. In the present study, the fail-safe coefficient was calculated using Comprehensive Meta-Analysis (CMA) as 1650 ($\alpha = 0.050$, $p = 0.000$), indicating that 1650 nonsignificant studies would need to be included to overturn the study findings. Taken together, the findings from the funnel plot, the loss-of-safety coefficient and the Begg and Egger tests collectively indicated the absence of significant bias in the literature included in this study.

Moderator analysis

Region

The impact of gamification on learning outcomes shows positive results across all regions, indicating that the utilization of gamification elements has a beneficial effect on students' learning outcomes irrespective of the region. The value of q represents the standardized sum of squares of effect sizes, and a higher q value corresponds to a smaller p-value, indicating greater heterogeneity among the studies. Table 4 reveals that the between-group effect of regional variables is significant, with $Q = 11.183$ and $p = 0.011 < 0.05$. This suggests that the effect of gamified chemistry teaching on learning outcomes differs significantly among regions. Due to the limited volume of literature available for America, this study primarily focuses on comparing the effect sizes observed in Asia and Europe. The experimental data clearly indicate that the larger effect values observed in Asia may be attributed to the varying levels of students across different regions within the continent. This finding aligns with the higher test scores reported by PISA for East Asian students in PISA 2018.

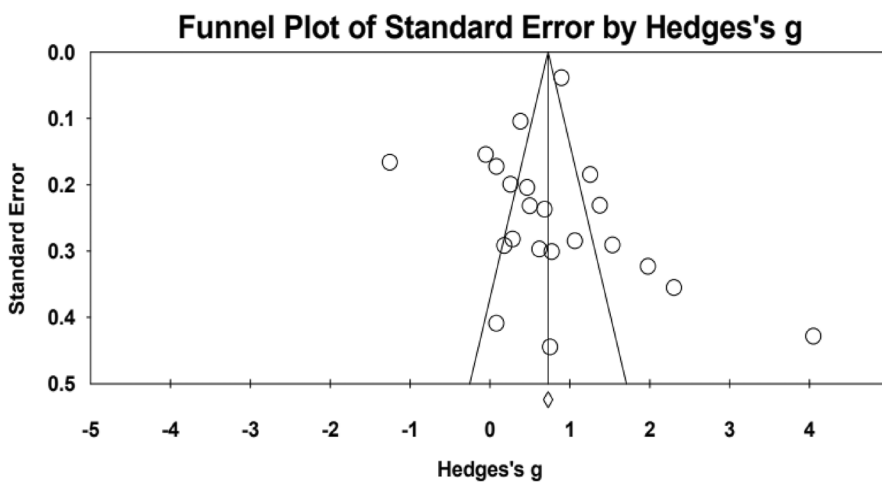

FIGURE 4 Publication bias.

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TABLE 4 The region effect size on students' achievement.

Moderator	*N*	Hedges' s g	*SE*	95%CI LL	95%CI UL	Overall effects
Asia	12	0.942	0.199	0.522	1.331	$Q=11.183$ $p=0.011<0.05$
Europe	7	0.512	0.386	−0.244	1.268	
America (N)	1	0.180	0.290	0.612	1.714	
America (S)	2	0.382	0.104	0.178	0.587	

TABLE 5 The educational level effect size on students' achievement.

Moderator	*N*	Hedges's g	*SE*	95%CI LL	95%CI UL	Overall effects
Elementary	6	1.195	0.482	0.251	2.139	$Q=2.621$ $p=0.454>0.05$
Highschool	3	0.930	0.374	0.197	1.664	
Middle	2	1.283	1.010	−0.697	3.262	
Undergraduate	11	0.476	0.241	−0.003	0.949	

Education level

This subgroup analysis categorized the school segments into four stages: elementary, middle school, high school and college. The results demonstrated a positive impact of gamification on learning effectiveness across all stages, with middle school showing the highest effect, followed by elementary school, high school and undergraduate levels (Table 5). The between-group effect, indicated by $Q=2.621$ and $p=0.454>0.05$, suggests that there is no significant difference in the effect of gamification application among school levels. In other words, the school level does not appear to be a determining factor influencing the effectiveness of gamification implementation. However, when considering the effect sizes for each school section, it can be observed that the implementation effect is stronger in both middle and high school levels compared to university and elementary schools.

Learning environment

The learning environment plays a significant role in influencing learning outcomes, with previous studies indicating that online and blended learning have a more positive impact on learning outcomes compared to face-to-face learning (Vallée et al., 2019). However, there are also studies suggesting that there may not be a significant difference in the impact on learning outcomes across the three environments. Therefore, in this study, the learning environment was examined as a moderating variable to explore its effect on the effectiveness of gamification. The results of the data analysis revealed positive effect values in all three teaching environments, namely online, offline and blended learning (Table 6). The between-group effect, represented by $Q=2.788$ and $p=0.248>0.05$, indicates that there is no significant difference in the effect of gamification on learning outcomes among the different teaching environments. In other words, the impact of gamification on learning effectiveness does not vary significantly depending on whether the learning is conducted online, offline or in a blended manner.

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TABLE 6 The learning environment effect size on students' achievement.

Moderator	*N*	Hedges's g	*SE*	95%CI LL	95%CI UL	Overall effects
Blended	6	0.279	0.406	−0.516	1.074	$Q=2.788$ $p=0.248>0.05$
Offline	10	1.150	0.333	0.497	1.802	
Online	6	0.742	0.179	0.391	1.093	

TABLE 7 The subject effect size on students' achievement.

Moderator	*N*	Hedges's g	*SE*	95%CI LL	95%CI UL	Overall effects
CS	11	0.501	0.253	0.006	0.997	$Q=11.943$ $p=0.036<0.05$
Economy	1	0.896	0.038	0.820	0.971	
Education	2	0.559	0.155	0.256	0.862	
Language	2	0.722	0.793	−0.832	2.276	
Management	1	1.255	0.185	0.894	1.617	
Science	5	1.519	0.620	0.303	2.375	

Types of subject

The positive impact of gamification on learning outcomes was observed across different disciplines, suggesting that the application of gamification elements has a beneficial effect regardless of the subject area. However, the effect of gamification application varied significantly between subjects, as indicated by the between-group effect of the subject variable (Table 7). The *Q* value of 11.754 and *p*-value of 0.038 ($p<0.05$) suggest that there are significant differences in the effect of gamification across different disciplines. Notably, the discipline of science exhibited the largest effect size, indicating that the implementation of gamification had a particularly strong positive impact in this subject area. This finding may be attributed to the specific characteristics and nature of science education and its unique approach to subject matter construction. These results emphasize that the effectiveness of gamification can be influenced by the subject being taught, and educators should consider tailoring gamification strategies to suit the specific requirements and characteristics of different disciplines.

Game elements

The findings presented in Table 8 indicate that, overall, the application of game elements has a positive effect on academic performance. However, it is important to note that one specific combination of game elements, namely Levels+Badges+Leaderboards, was found to have a negative effect on academic performance. This finding aligns with the results reported by Bai et al. (2019), further supporting the notion that this particular combination may not be conducive to optimal academic outcomes. Furthermore, the analysis revealed a significant difference in the effects of different game elements on academic achievement, as indicated by the *Q* value of 154.324 and *p*-value of 0.00 ($p<0.05$). Specifically, the combinations of Points+Badges+Leaderboards, Points+Leaderboards and Points+Badges+Leaderboards+Feedback exhibited higher effect values, suggesting that the inclusion of these

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TABLE 8 The game elements' effect size on students' achievement.

Moderator	*N*	Hedges's g	*SE*	95%CI LL	95%CI UL	Overall effects
Badges	1	0.257	0.199	−0.133	0.648	$Q=154.324$ $p=0.000<0.05$
Badges+Leaderboards	2	0.476	0.167	0.148	0.805	
Levels+Award+Badges+Leaderboards	1	−1.254	0.166	−1.579	−0.929	
Levels+Badges	2	0.515	0.168	0.185	0.845	
Levels+Points+Badges	1	0.180	0.292	−0.392	0.751	
None	3	1.203	0.367	0.483	1.923	
Points	2	2.153	1.883	−1.537	5.843	
Points+Badges+Feedback	1	0.081	0.409	−0.720	0.882	
Points+Badges+Leaderboards	5	0.842	0.320	0.214	1.470	
Points+Badges+Leaderboard+Feedback	1	2.304	0.355	1.608	3.000	
Points+Badges+Levels+Leaderboards	2	0.361	0.301	−0.228	0.950	
Points+Leaderboards	1	1.376	0.231	0.923	1.829	

game elements is associated with better academic results. However, it is important to acknowledge that the number of studies examining each type of game element was limited, which highlights the need for further comprehensive research to validate the findings of this study. Additional in-depth investigations are required to provide a more robust understanding of the effects of different game elements on academic performance.

DISCUSSION

The findings of the study encompass various aspects related to the impact of gamification on students' academic achievement. Initially, despite the minor disruption caused by Covid-19, there has been a notable increase in research activity within the gamification domain, indicative of its growing significance in educational discourse.

Our findings indicate a positive impact of gamification on academic performance, aligning with existing studies (Dehghanzadeh et al., 2021; Yıldırım & Şen, 2021). The effect of gamification on student achievement showcases a varied landscape encompassing different educational levels, subjects, learning environments, game elements and regions. Notably, the utilization of game elements such as points, badges and leaderboards emerge as prevalent strategies across studies, reflecting their efficacy in enhancing student engagement and motivation, as reported by other studies (Leitão et al., 2022). The efficacy of gamification in enhancing academic performance extends globally, with the most favourable results observed in Asia, followed by Europe. The moderator analysis, spanning region, education level, learning environment, subject and game elements, offers valuable insights into the differential effects of gamification across various educational parameters. Notably, while the positive impact of gamification is evident across different regions and education levels, the effectiveness of gamification strategies may vary based on factors such as subject area and choice of game elements (Legaki et al., 2020).

Our moderation analysis also revealed that the application of gamification in universities showed limited effectiveness. This might be attributed to factors such as reduced learning

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hours in universities, courses emphasizing practical aspects over academic performance and the diversified grading criteria where test scores constitute only a portion. For example, numerous college programming courses have transitioned from traditional exams to more code-writing exercises. Additionally, university learning places greater emphasis on autonomy and demands less teacher oversight, potentially diminishing the impact of gamification compared to primary, middle and high schools (Thomas et al., 2015).

The study underscores that gamification exerts a more pronounced impact in face-to-face learning environments compared to other settings, emphasizing its significance in such scenarios. This discrepancy is attributed to the limited interactivity inherent in online courses, resulting in decreased student engagement (Bastos et al., 2022; Shah et al., 2022). Notably, online learning outperformed blended learning, with the latter showcasing suboptimal effectiveness. Challenges in blended learning implementation, including resource allocation, teacher and student motivation and complex organization, may contribute to this.

Regarding the incorporation of game elements, the findings suggest that badges effectively incentivize students through reward systems (Shields & Chugh, 2017). This approach promotes focused learning rather than aimless gaming. While points and leaderboards may induce a sense of shame among lower-performing students, the study highlights the positive impact of leaderboards in fostering healthy competition, especially when only displaying the top scores while preserving the privacy of lower-scoring students. Notably, the study emphasizes that more game elements do not necessarily yield better results, cautioning against blind adoption and emphasizing the need for a strategic approach to maximize the positive impact of gamification on academic performance (Armier et al., 2016).

IMPLICATIONS AND FUTURE STUDY

This study contributes significantly to the existing literature in three key areas. First, it enhances our comprehension of gamification in educational environments by quantifying its impact on learners' academic performance, grounding its benefits in empirical evidence rather than unsubstantiated claims. Second, the research identifies moderating factors that influence the effectiveness of gamification. It underscores that the mere inclusion of gamification elements does not universally enhance learning outcomes; instead, it emphasizes the importance of tailoring these elements to individual learners' preferences and needs. The study reveals that younger students may benefit from straightforward game mechanics, immediate feedback and frequent rewards, while their college counterparts may prefer more complex games with progressive difficulty levels and social features for collaboration or competition. Notably, the study indicates that combinations such as Points, Points+Badges+Leaderboard+Feedback and Points+Leaderboards exhibit a more favorable effect size, suggesting their potential for improving students' learning outcomes. This finding offers guidance for educators to strategically incorporate these game elements in teaching practices. Additionally, it highlights the need for further exploration into different combinations of game elements and their impact on students' test scores. The study also underscores the influence of learners' characteristics, such as age, gender, prior knowledge and cognitive abilities, on the impact of gamification on academic performance. Considering educational level as a variable allows researchers to discern how gamification functions across diverse educational contexts, offering insights into effective practices and design principles tailored to each level. Lastly, educational institutions are urged to prioritize the enhancement of blended learning methodologies, ensuring they appeal to both educators and learners. This involves providing extensive resources and support to educators, facilitating their effective implementation of blended learning strategies. Additionally, institutions should strategically integrate gamification into these methodologies to maximize outcomes while minimizing resource expenditure.

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These findings enrich our comprehension of the intricate dynamics involved in implementing and assessing the effectiveness of gamification within educational environments. They serve as a catalyst for informed decision-making and inspire future research ventures in this field. Moving forward, it is imperative for gamification research to consider the unique dynamics present in student–teacher relationships, especially within university settings.

STUDY LIMITATIONS

This study has several limitations that should be acknowledged. Firstly, while the effectiveness of teaching encompasses both students' academic achievement and non-cognitive abilities, this study solely focuses on the effect of gamified instruction on academic achievement and does not consider the impact on non-cognitive abilities. It would be valuable for future research to explore the influence of gamified instruction on students' non-cognitive abilities as well. Another limitation is the reliance on a database-based literature search, omitting potential evidence from online research platforms, authoritative government reports and other non-traditional sources. This could introduce publication bias and selective reporting bias. Additionally, the study exclusively selected English literature, overlooking relevant research in other languages. Broadening the scope to include literature in various languages would enhance the comprehensiveness of the study.

CONCLUSIONS

Gamified learning has emerged as a widely embraced pedagogical theory and educational practice, recognized for its capacity to elevate motivation, enhance student engagement and ultimately bolster academic achievement (Zahedi et al., 2021). Rooted in the notion of harnessing students' intrinsic enjoyment of games to fuel their motivation for learning (Chiu et al., 2021), gamification has undergone extensive study, yielding varied perspectives on its impact on academic outcomes. While the existing literature showcases positive effects, a consensus remains elusive. This paper seeks to bridge this gap by offering a comprehensive overview of quantitative research in educational settings focused on gamification. Specifically, through a meta-analysis of 22 experimental studies conducted between 2008 and 2023, this research explored the influence of gamification on student academic performance, measured through objective instruments like test scores. The outcomes, analysed using a random effects model, revealed a moderately positive effect of gamification on academic performance. This study contributes empirical evidence affirming the positive influence of gamification on student academic performance. Additionally, the moderator analyses provide valuable insights into the strategic incorporation of game design elements and considerations tailored to diverse educational stages. As educators continue to explore innovative approaches, these findings serve as a guiding resource, offering practical considerations for the effective implementation of gamification in educational settings.

CONFLICT OF INTEREST STATEMENT

No potential conflict of interest was reported by the authors.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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ETHICS STATEMENT

This meta-analysis does not involve direct data collection from human participants.

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APPENDIX (SELECTED ARTICLES FOR META-ANALYSIS)

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